

Let's get back to the analysis of a Load Altering Attack.

To check the feasibility of a LAA, we have that $\forall(v, s)$:

- $P_v^{LG} = -K_{vs}^{LG} \cdot \omega_s$
- $K_{vs}^{LG, min}$ that makes LAA destabilising is determined with root locus.
- $P_{vs}^{LV} = -K_{vs}^{LG, min} \cdot \omega_s$. With this the load will start varying with increasing amplitude, and as for the frequency (with a set point of 60 Hz and safety limits given by 62 Hz and 58 Hz) we'll have some oscillations as well, with the oscillations going above/below the safety limits.

We can define $\omega_s^{max} = 2$ as the maximum error on the frequency. When the frequency goes above this value, the generator is disconnected. How much load does the attacker need to control to have a successful attack? (minimum value) It's given by $2 \cdot K_{vs}^{LG, min} \cdot \omega_s^{max}$. *I did not understand why.*

We compare this with the vulnerable load (the amount of power an attacker can control at a specific bus) at the bus. We divide the previous value with the available vulnerable load, and only when the value is < 1 ... *I got lost, it should be that when the value is smaller than one than the attacker can control the portion of the available load, so for example with 0.6 the attacker has to control 60%.*

As for the optimal protection, let's take $v \in \mathcal{V}$ and P_v^{LV} . We assume, for now, that the sensor bus s is known. One mean of protection is given by protecting a fraction of the total vulnerable load (so we need to decide, for each load, the amount of power load we want to protect). We call P_v^{LP} the amount of the vulnerable load to be protected at v .

Here are the goals:

1. Make destabilising Load Altering Attacks impossible.
2. Protect the least amount of load.

As this is an optimal protection, to solve this we need to solve an optimisation problem. As for the decision variables, we have:

- P_v^{LP} , this is the amount of vulnerable load at the generic victim bus v to be protected (to make multi-point attacks not successful)

$$\begin{aligned}
& \min \left\{ \sum_{v \in \mathcal{V}} P_v^{LP} \right\} \\
& \text{subject to} \\
& P_v^{LP} \leq P_v^{LV} \quad \forall v \in \mathcal{V} \\
& 2 \cdot K_{v,s}^{LG} \cdot \omega_s^{max} \leq P_v^{LV} - P_v^{LP} \quad \forall v \in \mathcal{V} \\
& (A - B \begin{bmatrix} 0 & 0 & K^{LG} \end{bmatrix}) \text{ stable}
\end{aligned}$$

The first constraint is telling that we cannot protect more than the maximum amount of vulnerable load.

With the second one, instead, we're making sure the attacker has all the power needed to follow the attack formula.

The last is the stability of the system, so the attack does not destabilise it.

As for the second constraint it could be seen as generous, but it's included to make sure the linear frame is still valid, with no saturation (??).

Let's focus on the third constraint now (stability constraint). In practice, one way of checking stability is given by taking a look at eigenvalues of $A' = (A - B \begin{bmatrix} 0 & 0 & K^{LG} \end{bmatrix})$. A problem now is that checking the eigenvalues is more of a procedure than a mathematical constraint. In fact, we compute $\det(A' - \lambda I) = 0$, find $\lambda_1, \dots, \lambda_n$ and then check $\text{Re}(\lambda_i) < 0 \forall i$. We thus need to find a way to write a mathematical constraint on stability to be embedded into the problem. For this, we use Lyapunov theory. In general, this is used for non-linear systems, but we're going to use it nevertheless.

Remember that stability is referred to so called "equilibrium points" of a system. Consider a system described by $\dot{x} = f(x)$. The equilibrium point x^{eq} is given by any state satisfying the condition that $f(x^{eq}) = 0$.

Given an equilibrium point x^{eq} , and we can assume that $x^{eq} = 0$, Lyapunov can be used to check stability. Let's check some preliminary definitions:

- Positive-definite functions. A function of the state $V(x) : \mathbb{R}^n \rightarrow \mathbb{R}$ (scalar function) is called positive-definite ($V > 0$) $\iff$:
 1. $V(0) = 0$;
 2. $V(x) > 0 \forall x \in B_0$, with B_0 being a region around the origin.

An example of V : Assume we have a system with two state variables, we could have $V(x) = 1x_1^2 + 2x_2^2 = \begin{pmatrix} x_1 & x_2 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = x^T M x$. (quadratic form)

In general $V(x) = x^T M x$ is positive-definite $\iff$ matrix M has all eigenvalues positive. We say $V > 0 \iff M > 0$.

- Negative-definite functions. V is negative-definite $\iff (-V > 0)$.
- Semi positive-definite functions. V is S.P.D. $(V \geq 0) \iff$
 1. $V(0) = 0$
 2. $V(x) \geq 0 \forall x \in B_0$
- Semi negative-definite functions. V is S.N.D. $(V \leq 0) \iff (-V \geq 0)$

Let's give a visual interpretation of $V > 0$, with $n = 2$.

We have the plane with x_1 and x_2 , and as a "third axis" we sketch $V(x)$. We'll have a sort of "bowl" starting from the origin of (x_1, x_2) , and then grows with $V(x)$

✓ Lyapunov Theorem for Asymptotic Stability

The equilibrium $x^{eq} = 0$ of system $\dot{x} = f(x)$ is asymptotically stable if $\exists$ a scalar function of the state $V(x)$ that satisfies two conditions:

1. $V > 0$;
2. $\frac{d}{dt}V = \dot{V}$ must be < 0 (strictly negative-definite).

We need to find a V that satisfies the two conditions. Let's focus on the second one. We have that $\dot{V}$ is the variation of V (the derivative of V) along the trajectories of the system.

If we initialise the system at the equilibrium point, the state would not change. If we initialise the system a bit far away from the equilibrium point, the system will move. We can draw the trajectory of the system.

$\dot{V}(x(t)) < 0$ means that $V(x(t))$ decreases as t goes to zero.